Logistic Regression

1. Logistic regression is primarily used for:
   A. Regression problems
   B. Classification problems
   C. Clustering problems
   D. Dimensionality reduction

2. The output of logistic regression represents:
   A. Continuous value
   B. Probability
   C. Clusters
   D. Raw labels

3. The function used in logistic regression is:
   A. Linear function
   B. Sigmoid function
   C. Step function
   D. Gaussian function

4. The sigmoid function outputs values between:
   A. -∞ to +∞
   B. 0 to 1
   C. -1 to 1
   D. 1 to 100

5. Logistic regression assumes relationship between features and:
   A. Probability of class
   B. Cluster centers
   C. Variance
   D. Distance metric

6. Decision boundary in logistic regression is:
   A. Always curved
   B. Always linear
   C. Always circular
   D. Random

7. If predicted probability is 0.7, predicted class (threshold 0.5) is:
   A. 0
   B. 1
   C. Undefined
   D. Both

8. Logistic regression is based on:
   A. Least squares
   B. Maximum likelihood estimation
   C. K-means optimization
   D. Gradient boosting

9. Which loss function is used in logistic regression?
   A. MSE
   B. Cross-entropy loss
   C. Hinge loss
   D. Absolute error

10. Logistic regression is best suited for:
    A. Continuous output prediction
    B. Binary classification
    C. Image generation
    D. Clustering

11. A probability output of 0.49 means:
    A. Class 1
    B. Class 0
    C. Certain prediction
    D. Invalid output

12. If threshold is increased from 0.5 to 0.8, model becomes:
    A. More strict in predicting class 1
    B. More sensitive to positives
    C. Unchanged
    D. Random

13. Logistic regression works best when:
    A. Data is non-linear and chaotic
    B. Classes are linearly separable
    C. Data is unstructured
    D. Features are images

14. Multicollinearity in logistic regression affects:
    A. Predictions only
    B. Coefficient stability
    C. Data size
    D. Class labels

15. If model outputs only 0 or 1 always, it may indicate:
    A. Perfect model
    B. Overconfidence or overfitting
    C. Good calibration
    D. High bias always

16. Odds in logistic regression refer to:
    A. Probability ratio
    B. Feature weights
    C. Error rate
    D. Distance measure

17. Log-odds is also called:
    A. Sigmoid
    B. Logit
    C. Softmax
    D. Entropy

18. Increasing regularization in logistic regression:
    A. Increases overfitting
    B. Reduces overfitting
    C. Removes data
    D. Increases accuracy always

19. Which metric is NOT ideal for imbalanced data?
    A. Accuracy
    B. F1-score

C. Precision
D. Recall

20. Logistic regression outputs can be interpreted as:
A. Distances
B. Probabilities
C. Clusters
D. Eigenvalues

21. What does this code do?

"model.fit(X_train, y_train)"

A. Predicts outputs
B. Trains logistic model
C. Evaluates model
D. Normalizes data

22. What is returned by:

"model.predict(X_test)"

A. Probabilities
B. Class labels
C. Loss values
D. Feature importance

23. What does this return?

"model.predict_proba(X_test)"

A. Only class 1
B. Probabilities of each class
C. Labels only
D. Errors

24. Which function is used in sklearn for logistic regression?
A. LinearRegression()
B. LogisticRegression()
C. ClassifierLinear()
D. SigmoidRegression()

25. What happens if you skip scaling in logistic regression?
A. Always improves performance
B. May affect convergence
C. Converts to regression
D. No impact ever

26. What does this represent?

"model.coef_"

A. Intercept
B. Feature weights
C. Predictions
D. Loss

27. What does this represent?

"model.intercept_"

A. Feature importance
B. Bias term
C. Accuracy
D. Gradient

28. Which is true about .predict()?
A. Returns probabilities
B. Returns class labels
C. Returns features
D. Returns loss

29. What does this compute?

"log_loss(y_true, y_pred_proba)"

A. Accuracy
B. Cross entropy loss
C. Clustering score
D. Regression error

30. What happens if target is continuous in logistic regression?
A. Works normally
B. Becomes regression problem
C. Fails or produces incorrect output
D. Improves accuracy

31. Which library is commonly used?
A. sklearn
B. pandas only
C. matplotlib
D. seaborn

32. What does regularization control?
A. Data size
B. Model complexity
C. Number of rows
D. Output format

33. What does C parameter in logistic regression control?
A. Learning rate
B. Regularization strength
C. Number of classes
D. Feature count

34. What happens if C is very large?
A. Strong regularization
B. Weak regularization
C. No training
D. Random output

35. What is effect of solver in logistic regression?
A. Changes dataset
B. Optimization method
C. Feature selection
D. Output scaling

36. If model performs well on training but poorly on test data:
A. Underfitting

B. Overfitting
C. Perfect model
D. No learning

37. If dataset is highly imbalanced, best metric is:
A. Accuracy
B. Precision/Recall
C. MSE
D. R^2

38. Logistic regression fails when:
A. Data is linearly separable
B. Data is highly non-linear
C. Features are scaled
D. Data is clean

39. If decision boundary is straight line, model is:
A. Logistic regression
B. Neural network
C. KNN
D. Decision tree

40. Adding irrelevant features leads to:
A. Better accuracy always
B. Noise in model
C. Lower bias always
D. Perfect separation

41. If probability outputs are all ~0.5:
A. Model is confident
B. Model is uncertain
C. Model is perfect
D. Model is overfitting

42. If threshold is reduced to 0.2:
A. Fewer positives
B. More positives
C. No change
D. Model breaks

43. If data is perfectly separable, logistic regression:
A. Fails completely
B. Works well but may overfit
C. Becomes clustering
D. Cannot be used

44. Regularization helps in:
A. Increasing features
B. Reducing overfitting
C. Increasing noise
D. Changing dataset

45. If coefficients are extremely large, it indicates:
A. Underfitting
B. Overfitting
C. Perfect model
D. No issue

46. Logistic regression is a:
    A. Non-parametric model
    B. Parametric model
    C. Clustering model
    D. Tree-based model

47. If precision is high but recall is low:
    A. Many false negatives
    B. Many false positives
    C. Perfect model
    D. Balanced model

48. If dataset has outliers, logistic regression:
    A. Unaffected
    B. Sensitive
    C. Always improves
    D. Converts to clustering

49. Best way to improve logistic regression is:
    A. Remove features randomly
    B. Feature engineering
    C. Increase noise
    D. Reduce data size

50. Logistic regression outputs are transformed using:
    A. Softmax only
    B. Sigmoid
    C. ReLU
    D. Step function